

DISTRIBUCIÓN DE ESPECIES Y CAMBIO CLIMÁTICO

Dia 3 – Intercambio de calor y masa Sensibilidad térmica del organismo

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Dos aspectos importantes de la biología de los anfibios

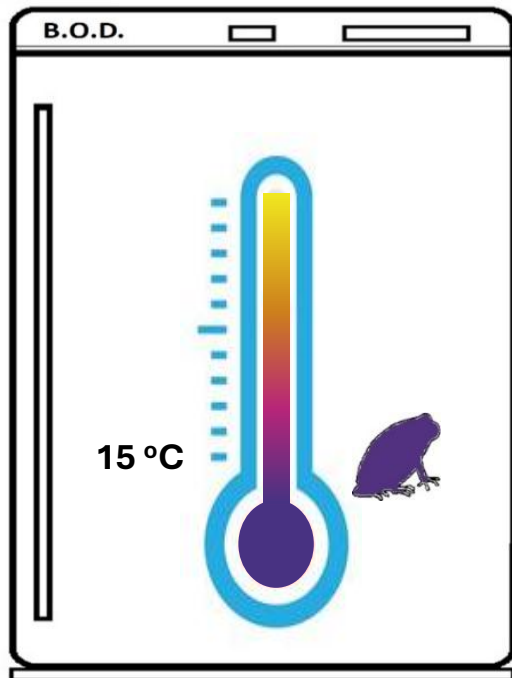
PIEL ALTAMENTE PERMEABLE

Baja resistencia de la piel resulta en altas tasas de pérdida evaporativa de agua

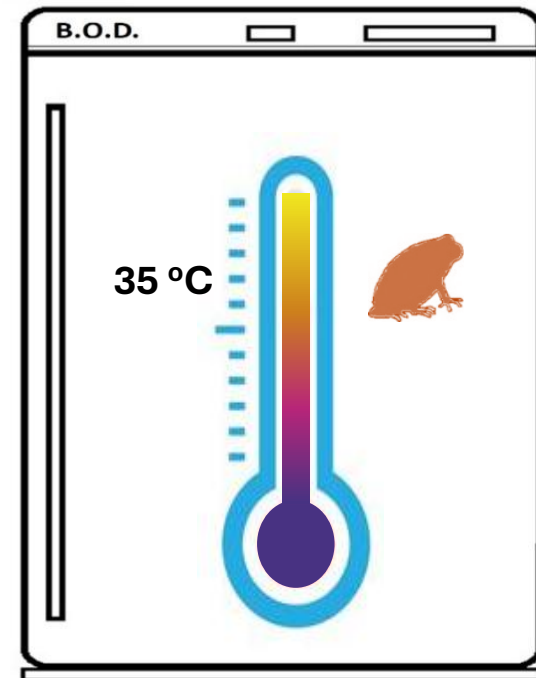
ANIMALES ECTOTÉRMICOS

Temperatura corporal altamente influenciada por la temperatura ambiental

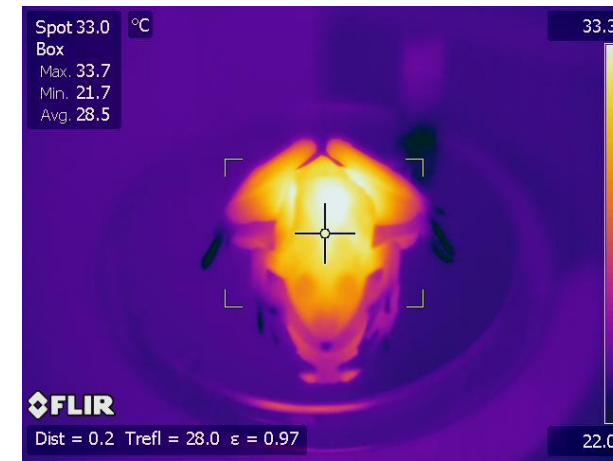
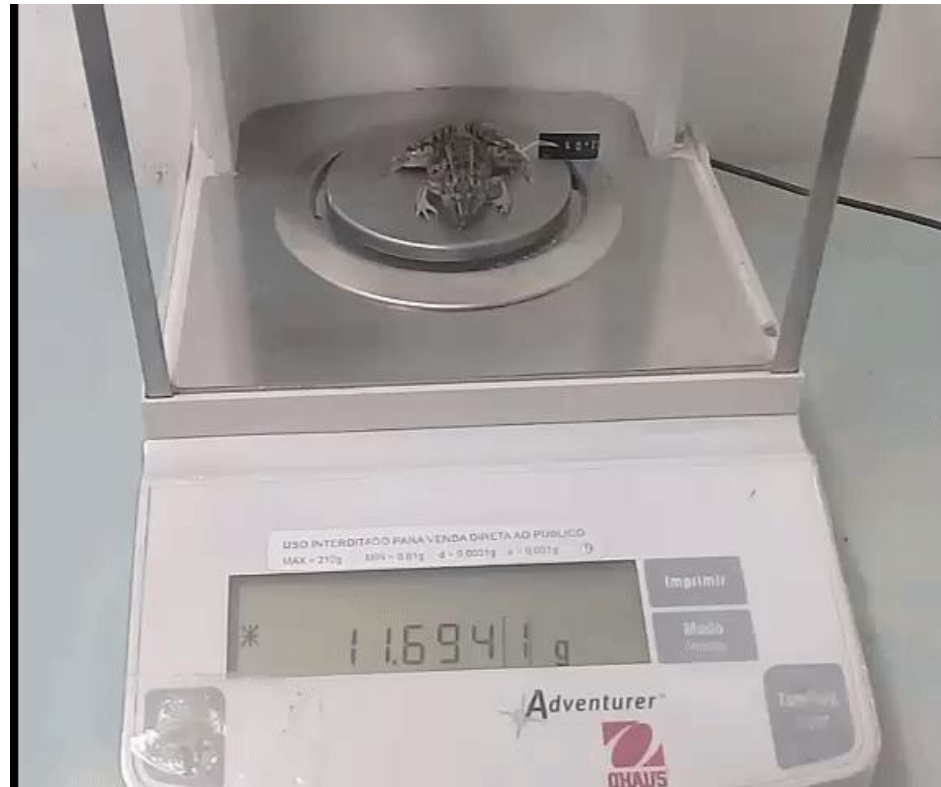
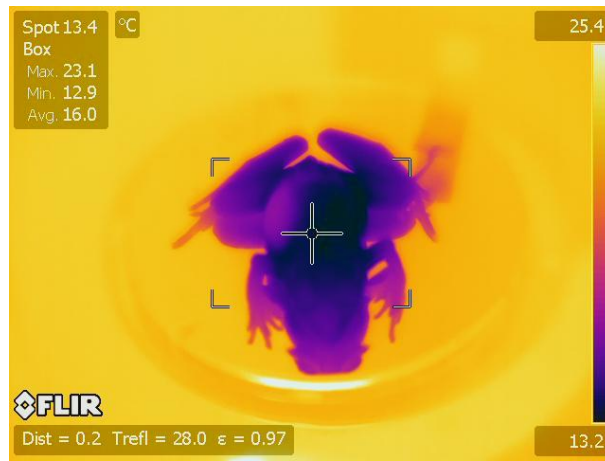
15 °C



35 °C



15 °C



30 °C



Dos aspectos importantes de la biología de los anfibios

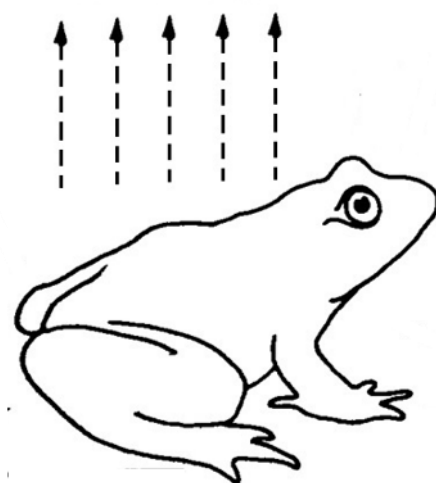
PIEL ALTAMENTE PERMEABLE

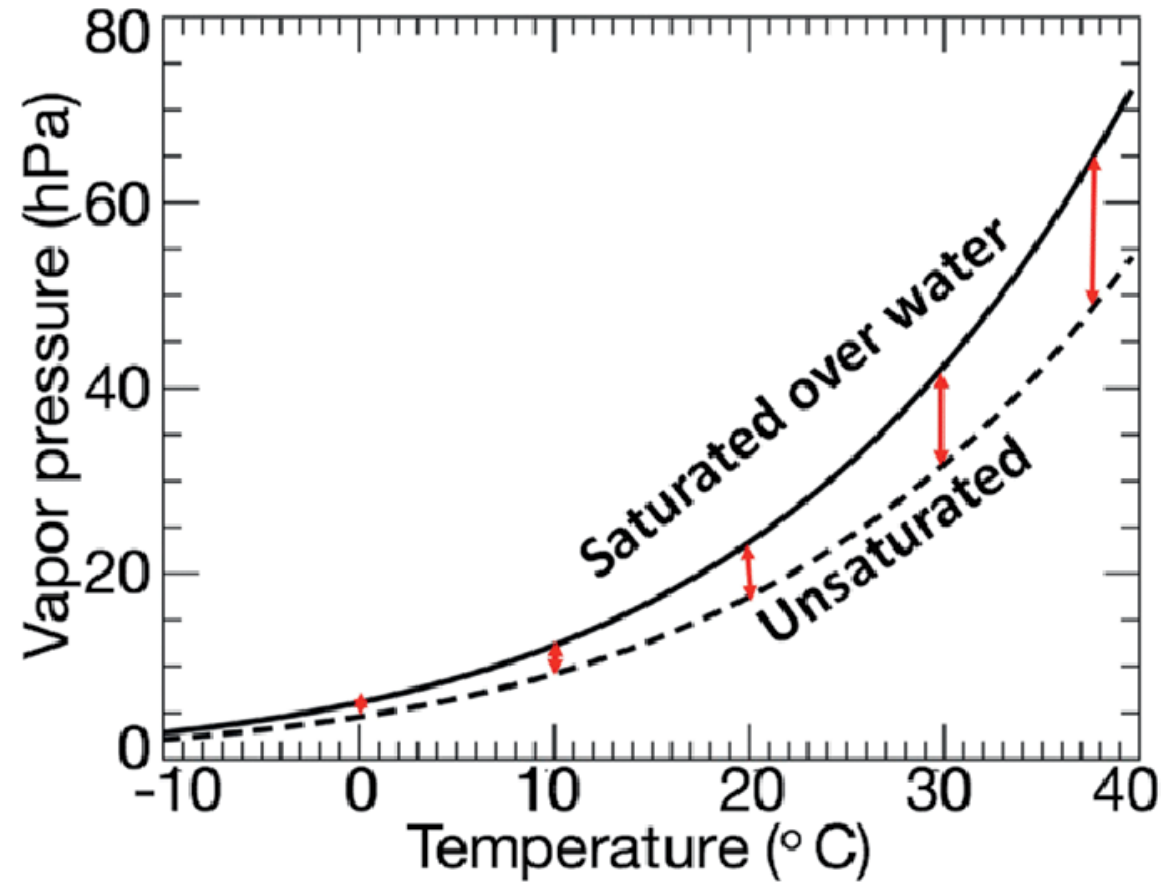
Baja resistencia de la piel resulta en altas tasas de pérdida evaporativa de agua

ANIMALES ECTOTÉRMICOS

Temperatura corporal altamente influenciada por la temperatura ambiental

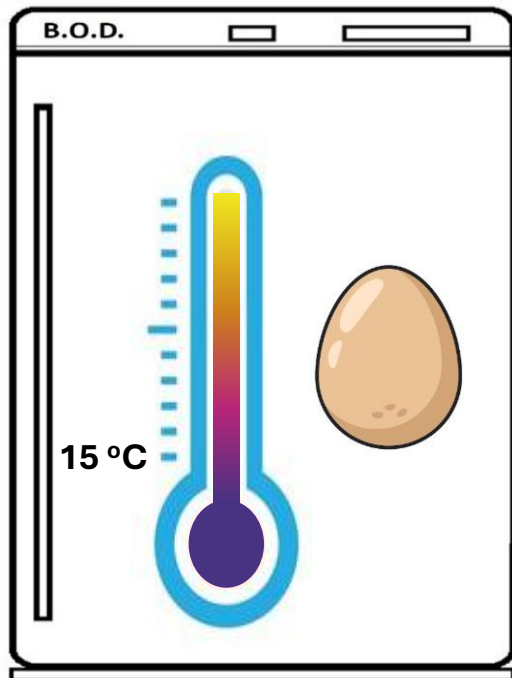
Pérdida de calor y masa corporal



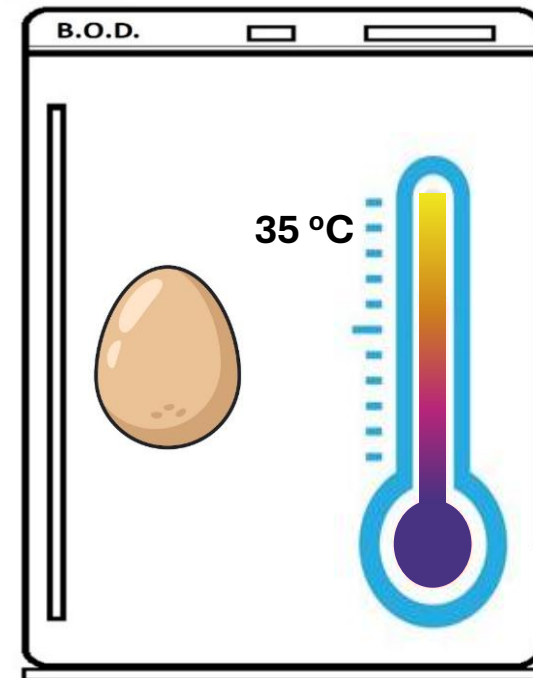


Al subir la temperatura, aumenta el déficit de presión de vapor (DPV). Esto genera una "sed" ambiental que acelera drásticamente la evaporación en superficies húmedas (e.g., la piel de um anfibio)

15 °C



35 °C



15 °C



35 °C

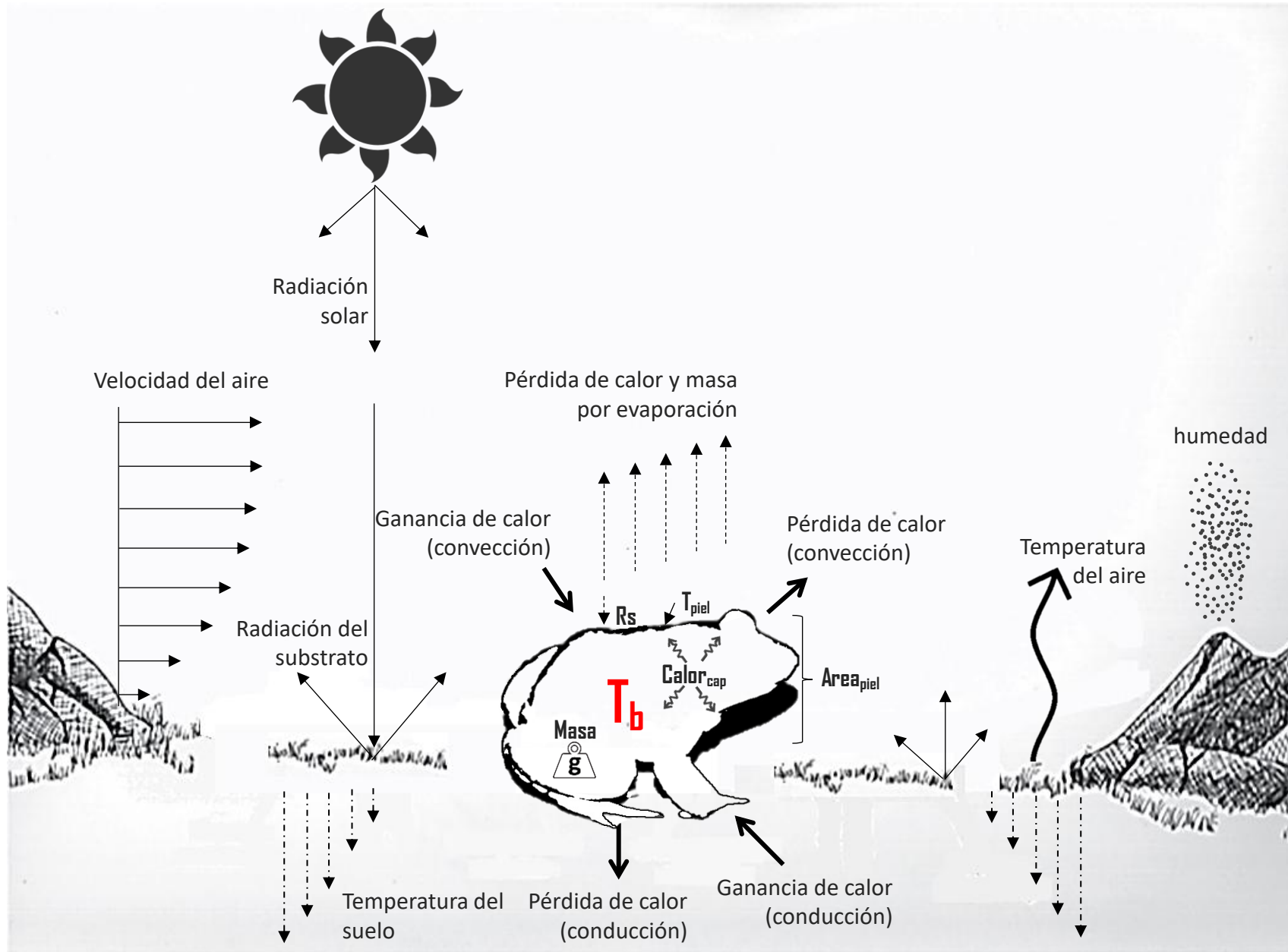


Otros factores que influyen en el ganancia y pérdida de agua

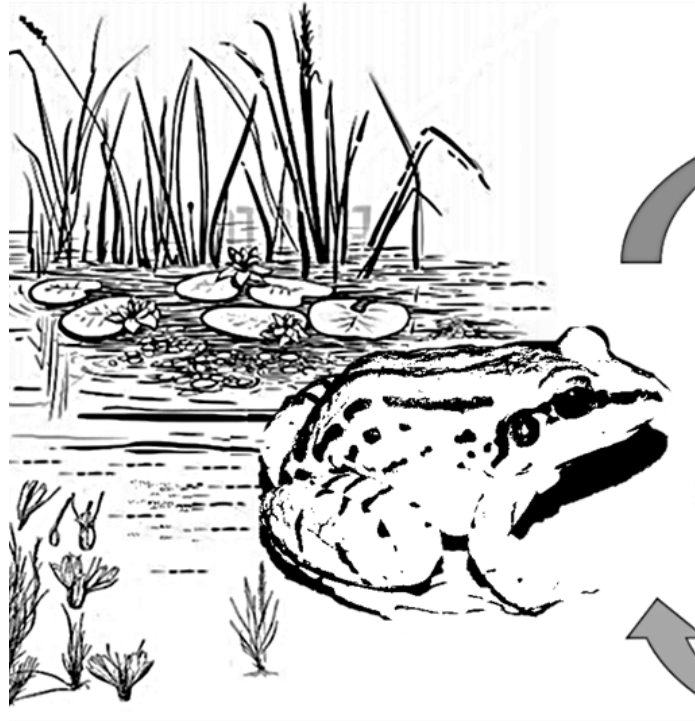
Masa Corporal (Masa vs. Área): Individuos pequeños tienen mucha más piel expuesta en relación a su volumen corporal que individuos más grandes.

Resistencia cutánea a la evaporación: Algunas especies de anfibios poseen secreciones cerosas que disminuyen bastante la permeabilidad de la piel, consecuentemente reduciendo la pérdida de agua por evaporación.

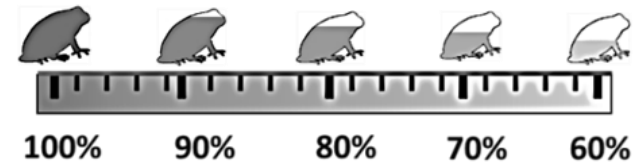
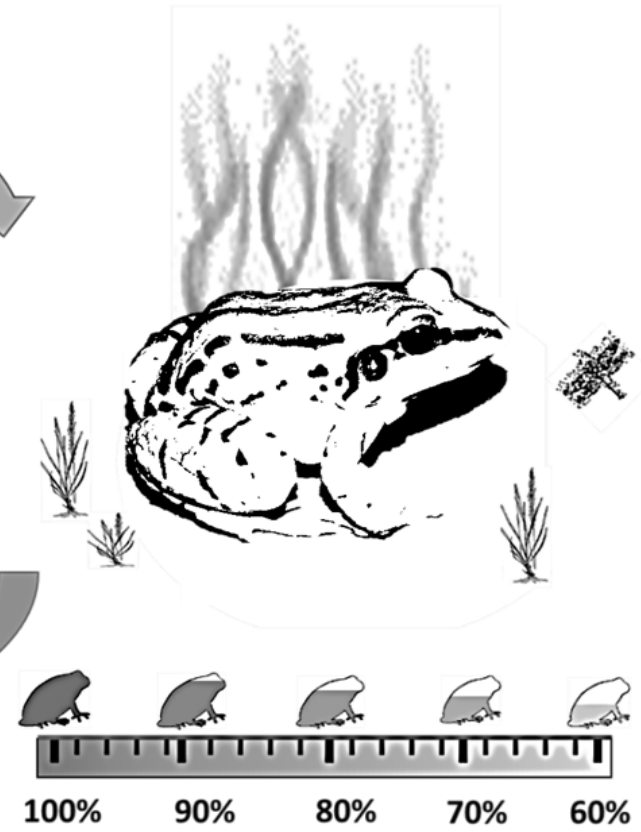
Viento: El viento fuerte "barre" la capa protectora de aire húmedo que se encuentra casi adherido a toda superficie húmeda (e.g., la piel del anfibio).



Hidratación



Deshidratación



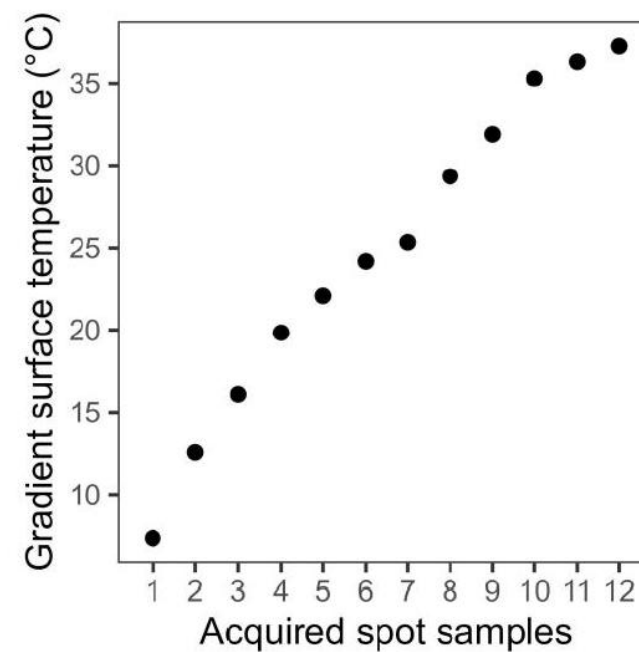
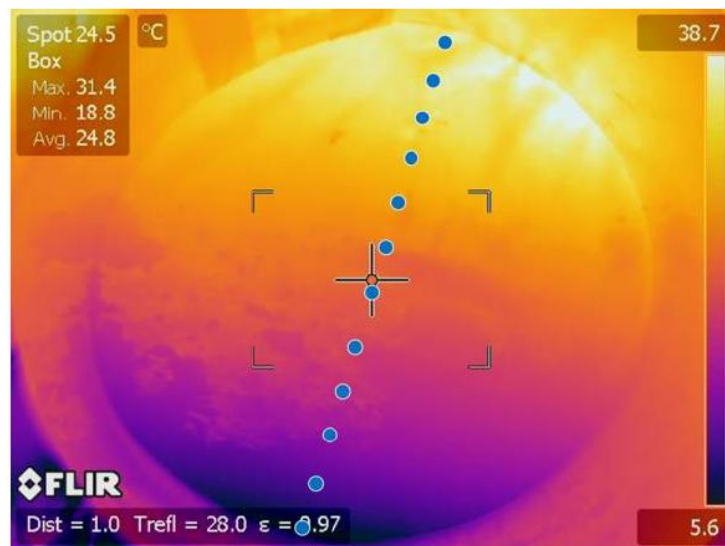
Condiciones térmicas preferida por la especie?



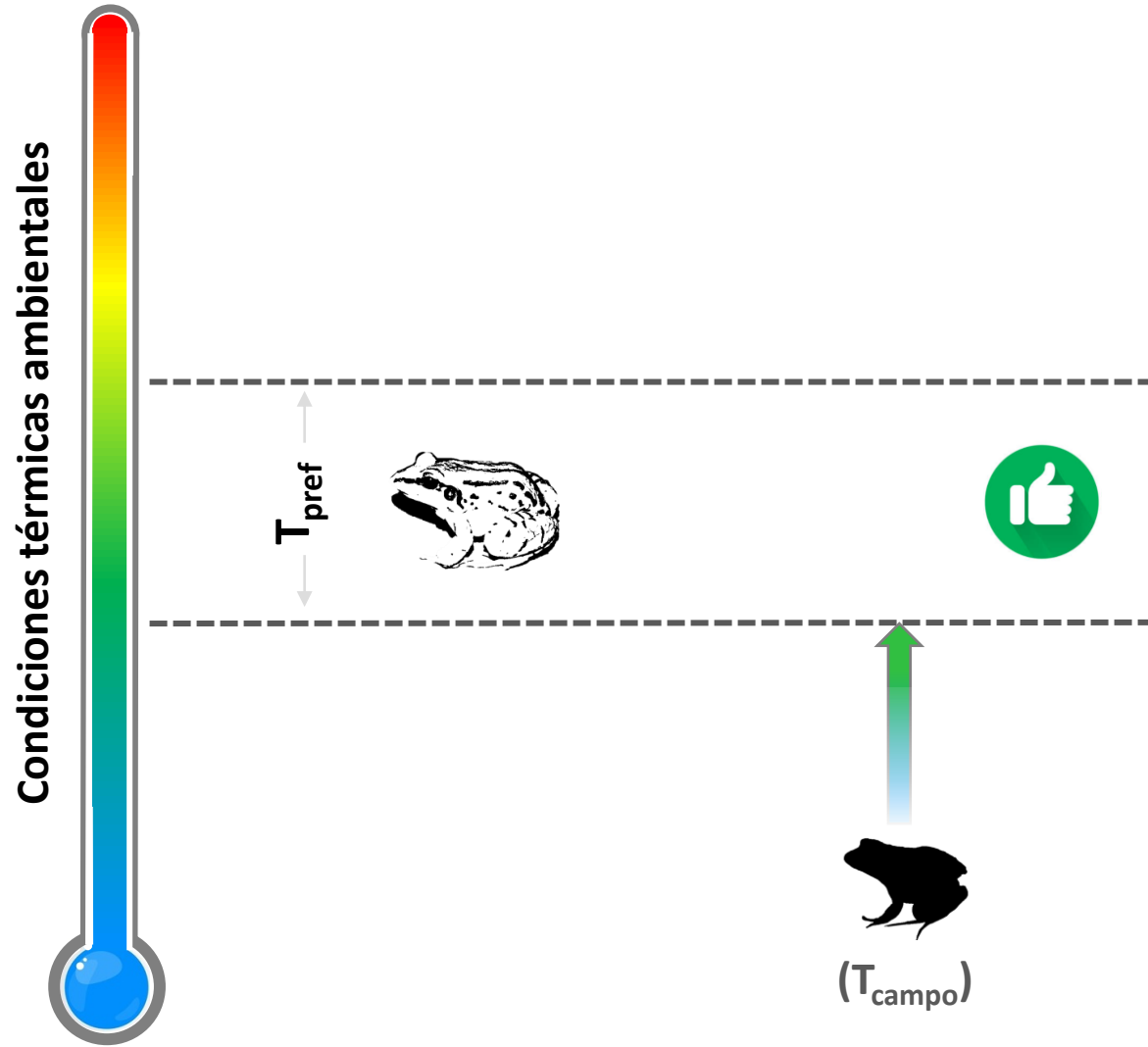


Temperatura
corporal en campo
(T_{campo})

Gradiente térmico

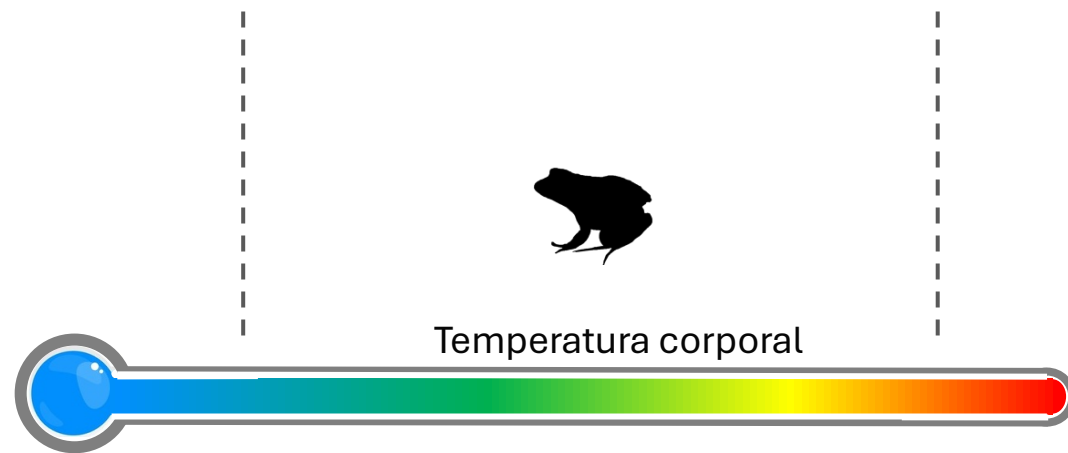


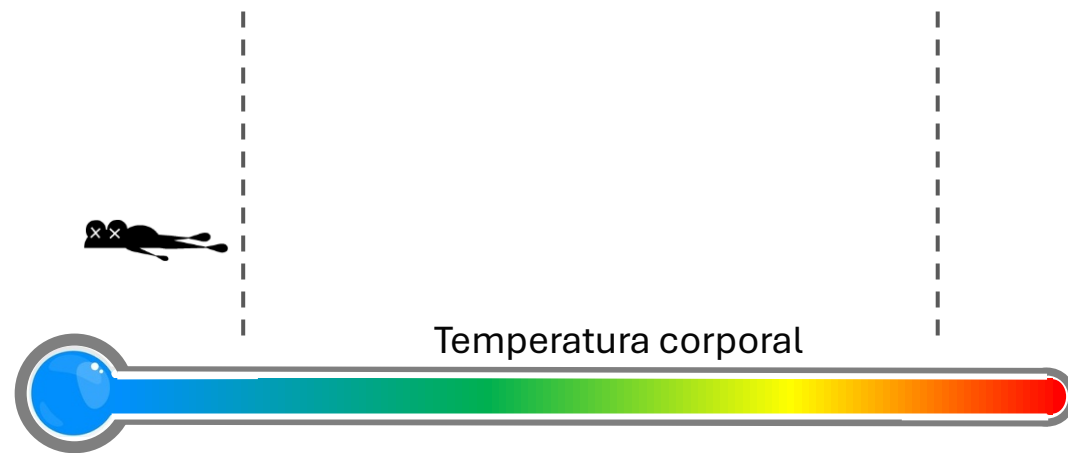
Utilizado para determinar la temperatura preferida (**T_{pref}**) de la especie em condiciones de 0 costos de termoregulación

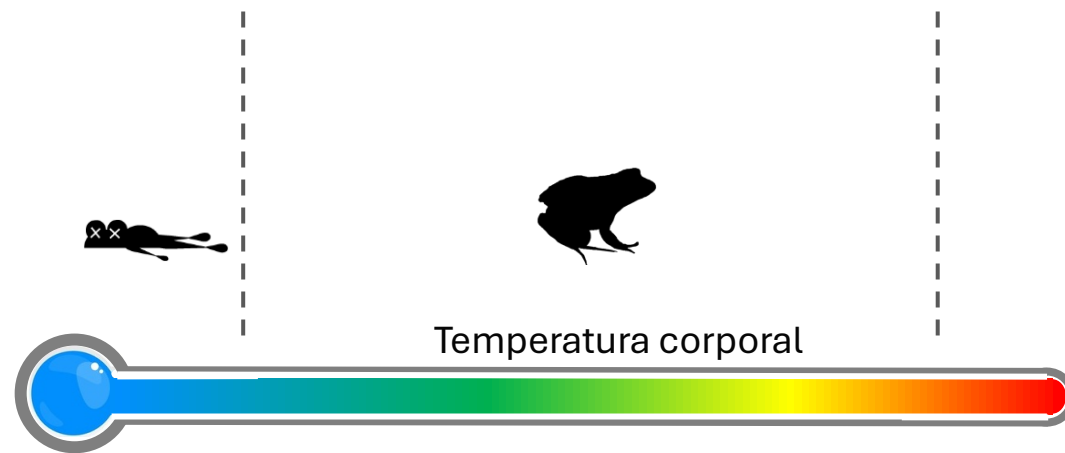


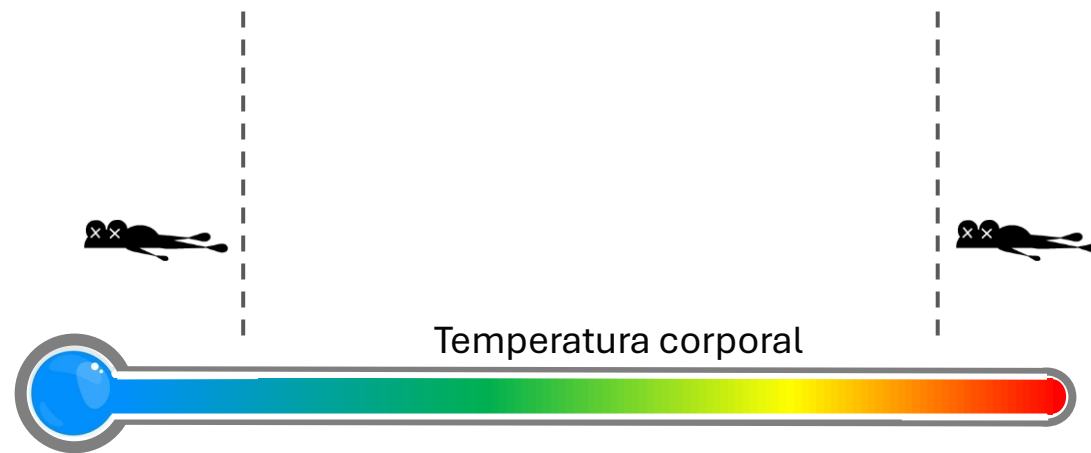
Condiciones térmicas ambientales que permitan a la especie alcanzar su temperatura preferida se presume que son ambientes térmicamente adecuados para la especie.

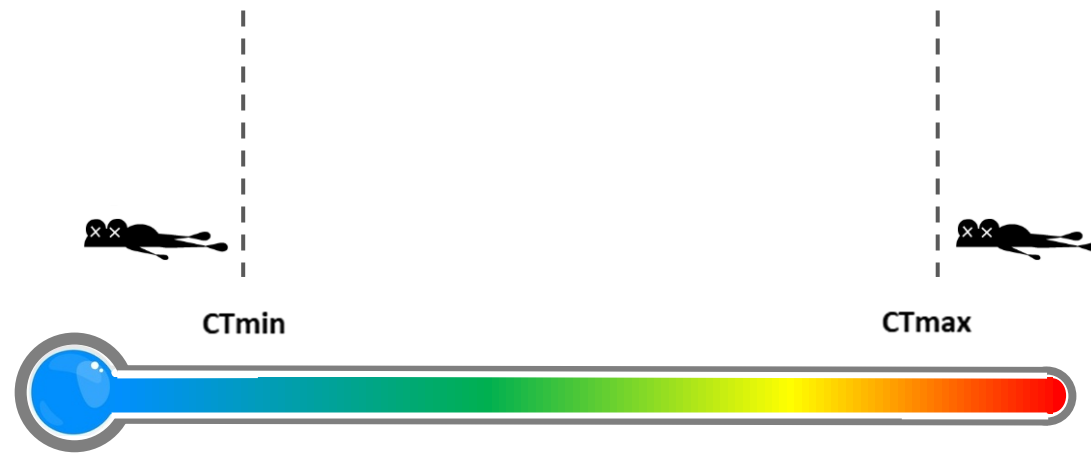
La relación de la temperatura y el desempeño del organismo

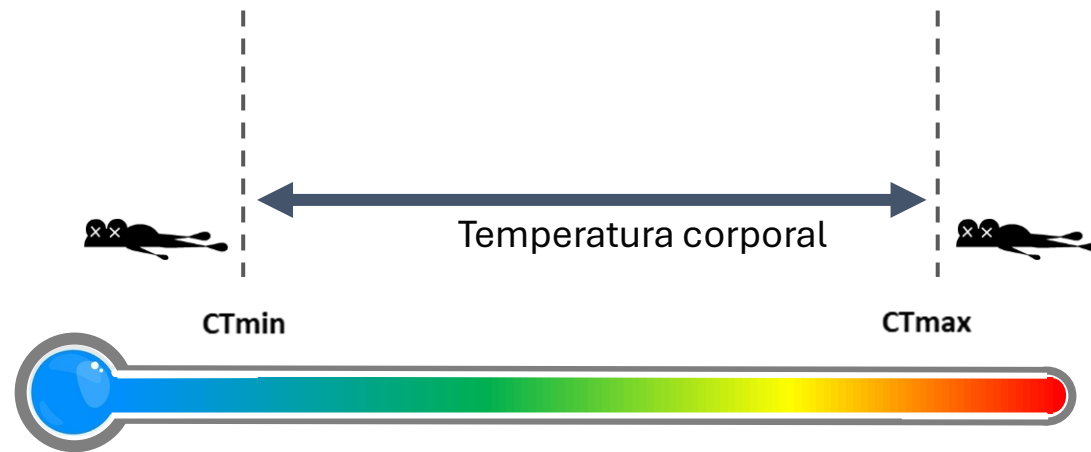




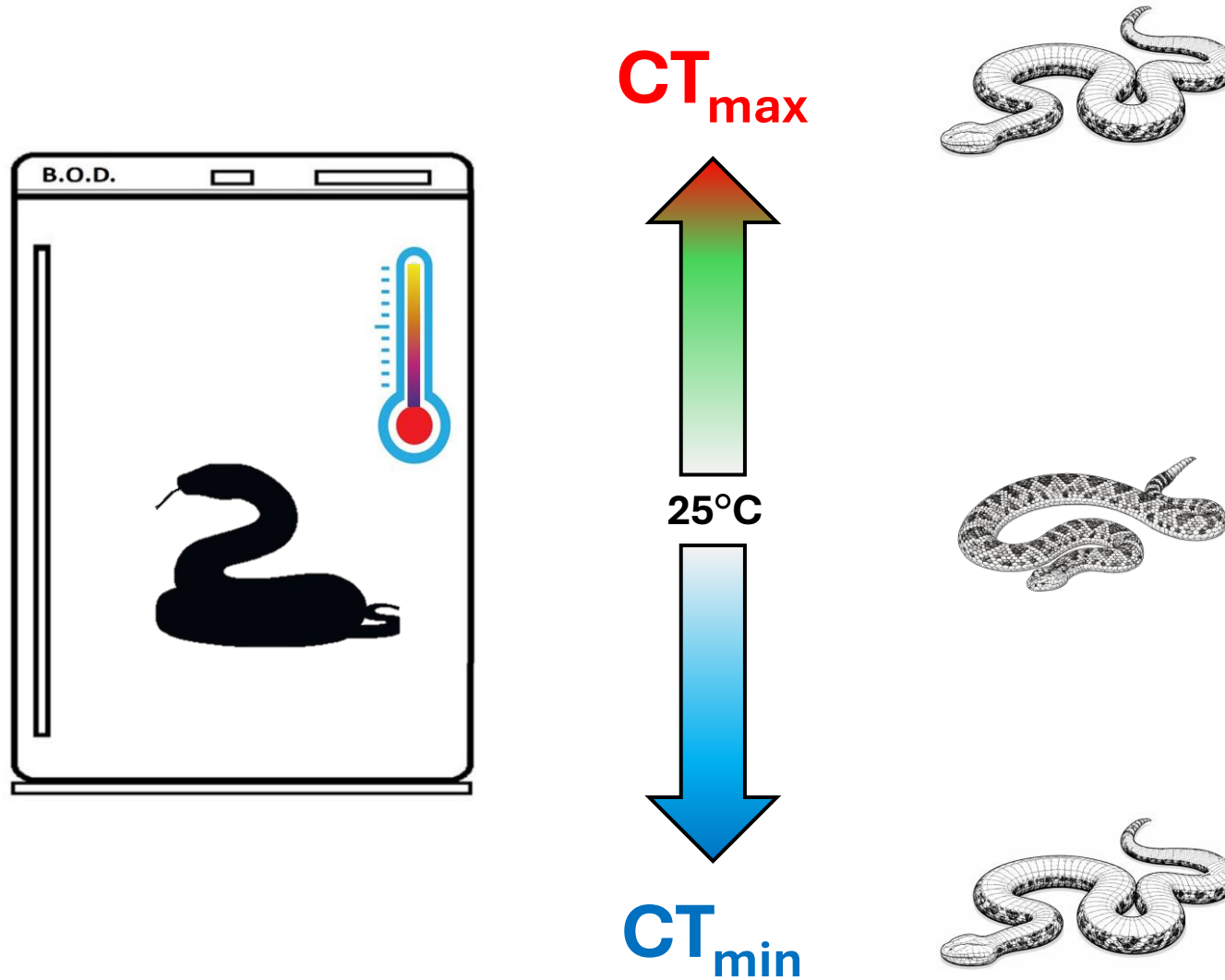


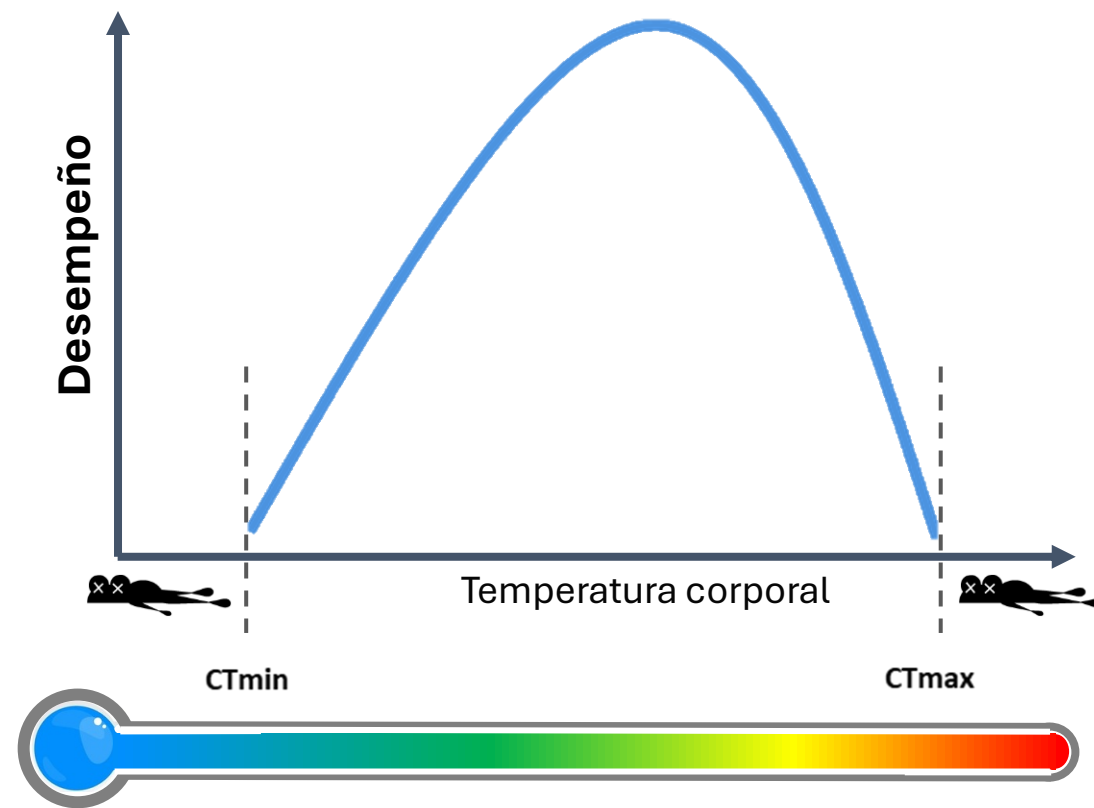


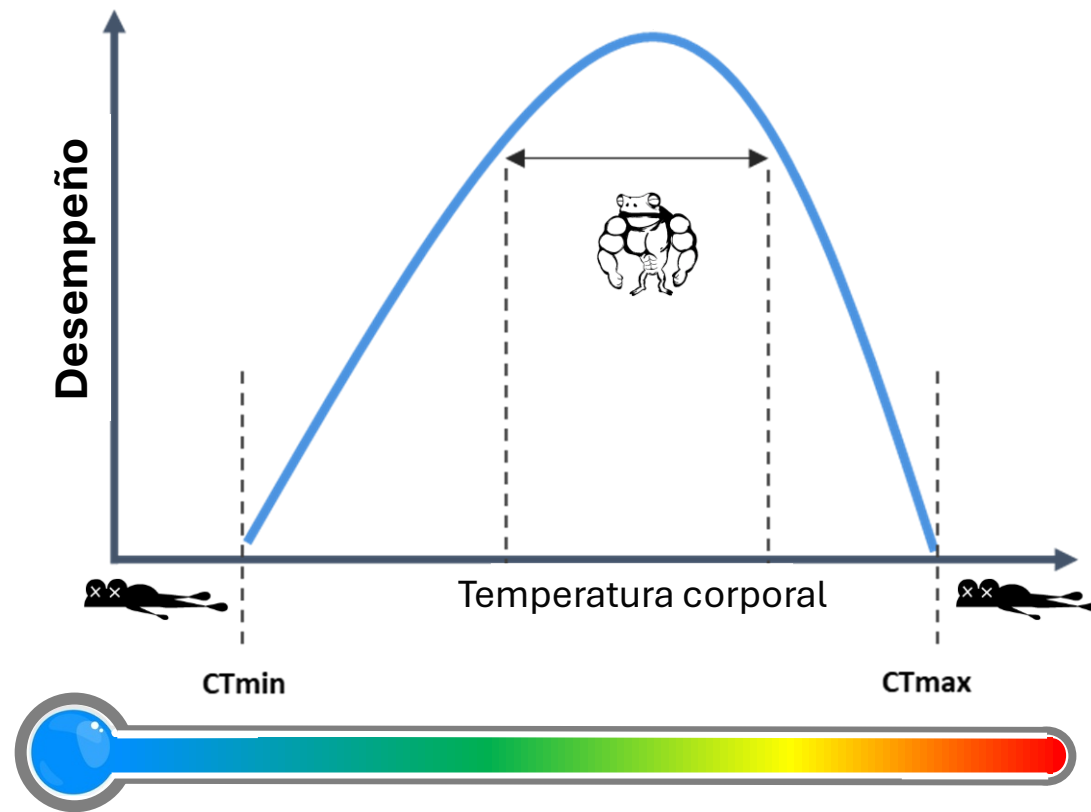


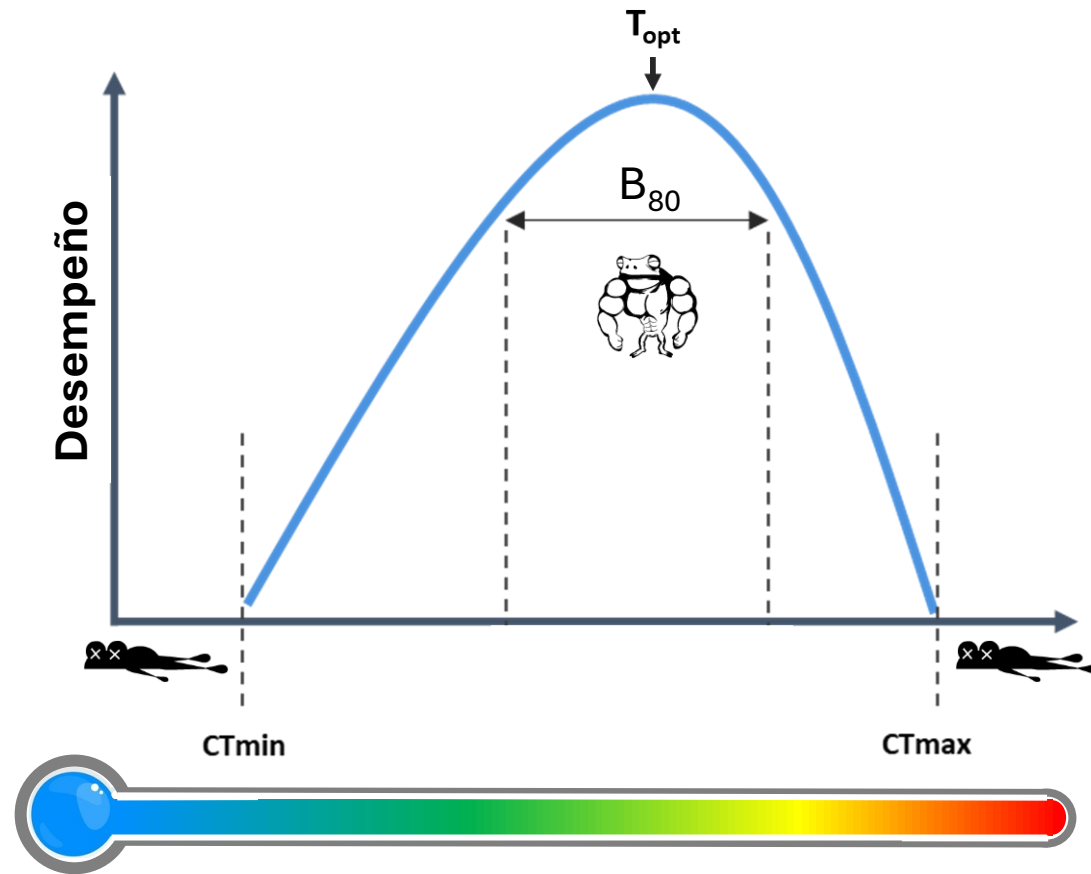


TOLERÂNCIAS TÉRMICAS









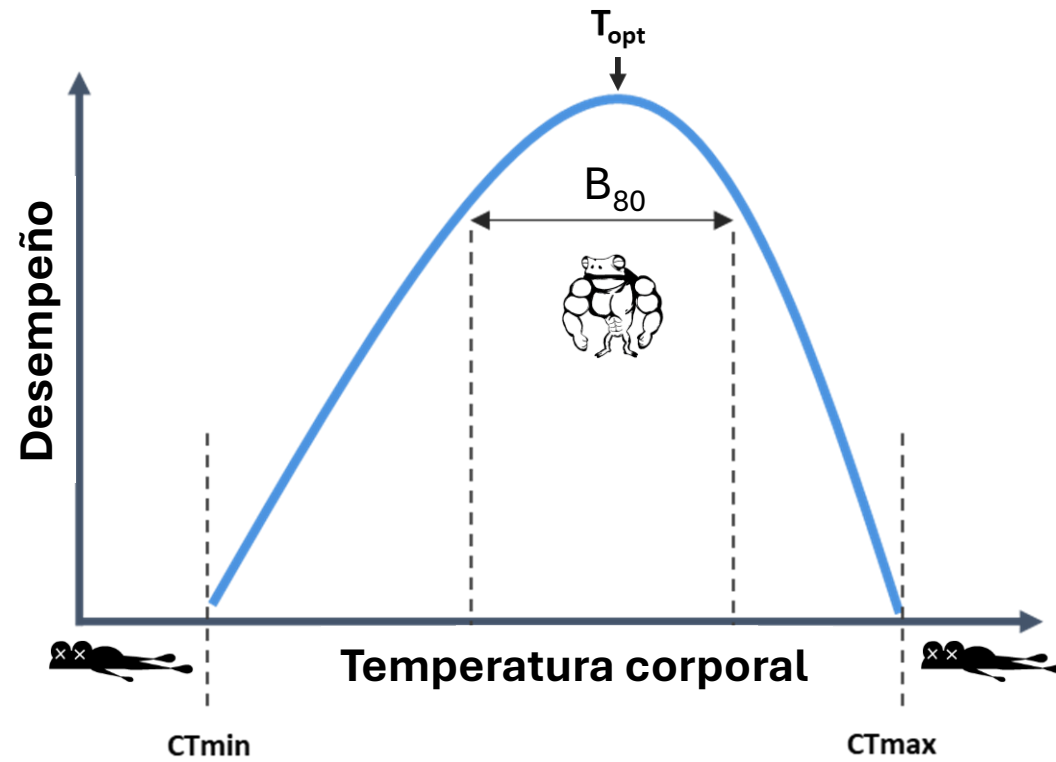
Temperatura óptima (T_{opt})

La temperatura corporal en la que se alcanza el desempeño máximo

Amplitud de desempeño (B_{80})

La faja de temperaturas corporales en la que podemos mantener al menos el 80% del máximo desempeño

CURVA DE DESEMPEÑO TÉRMICO





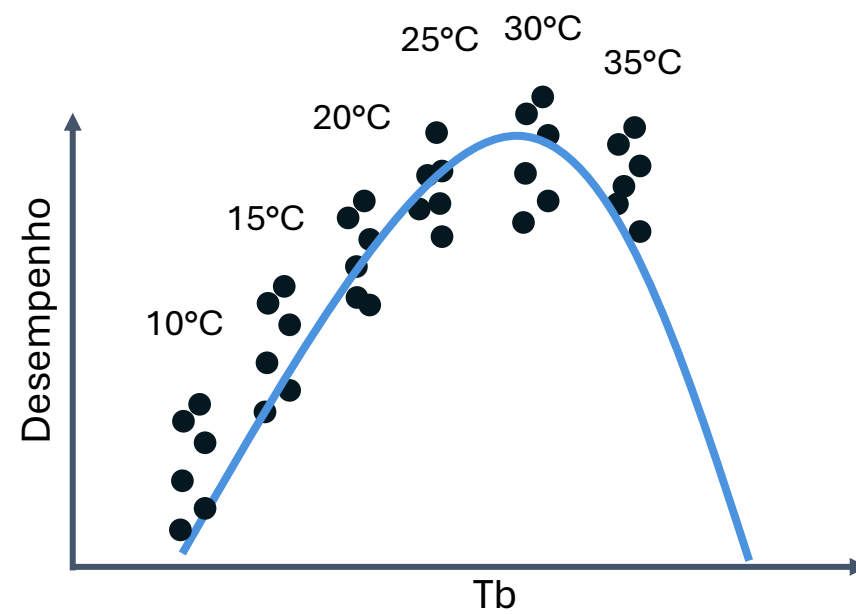




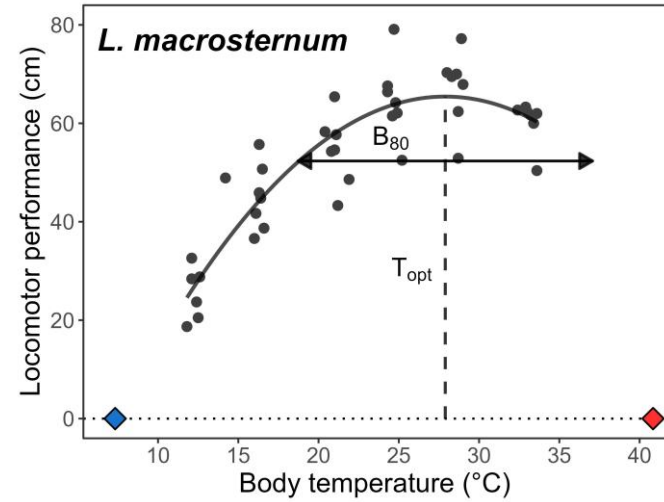
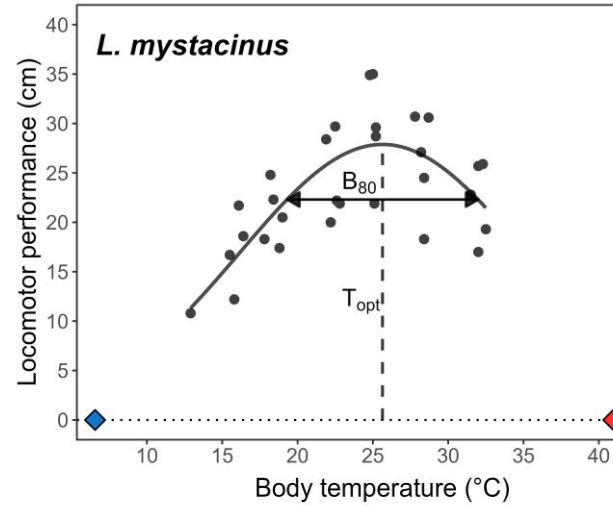
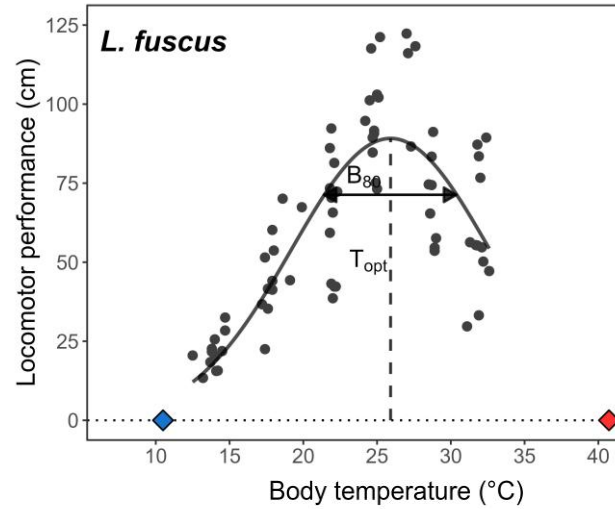


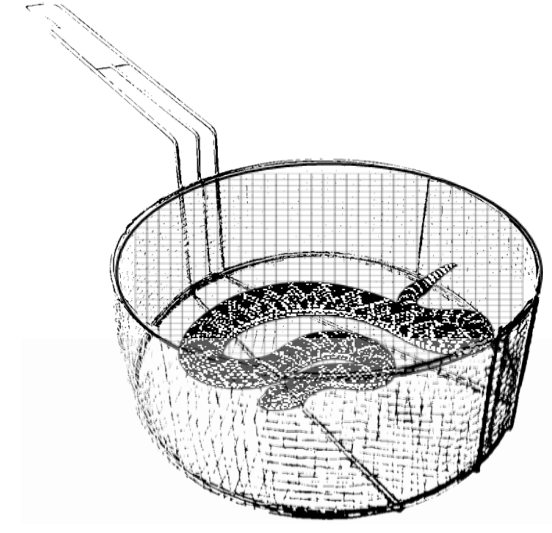
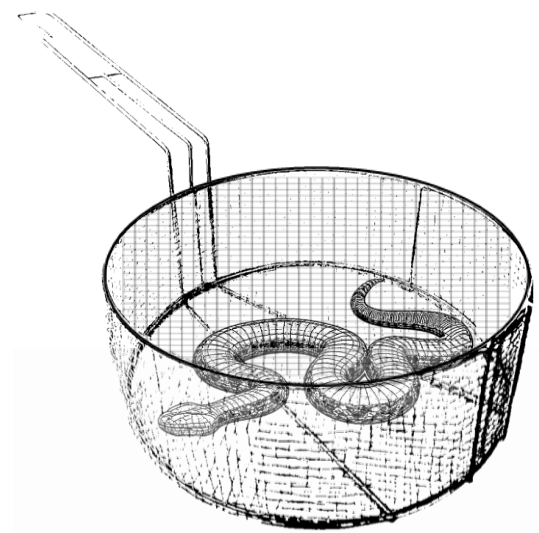
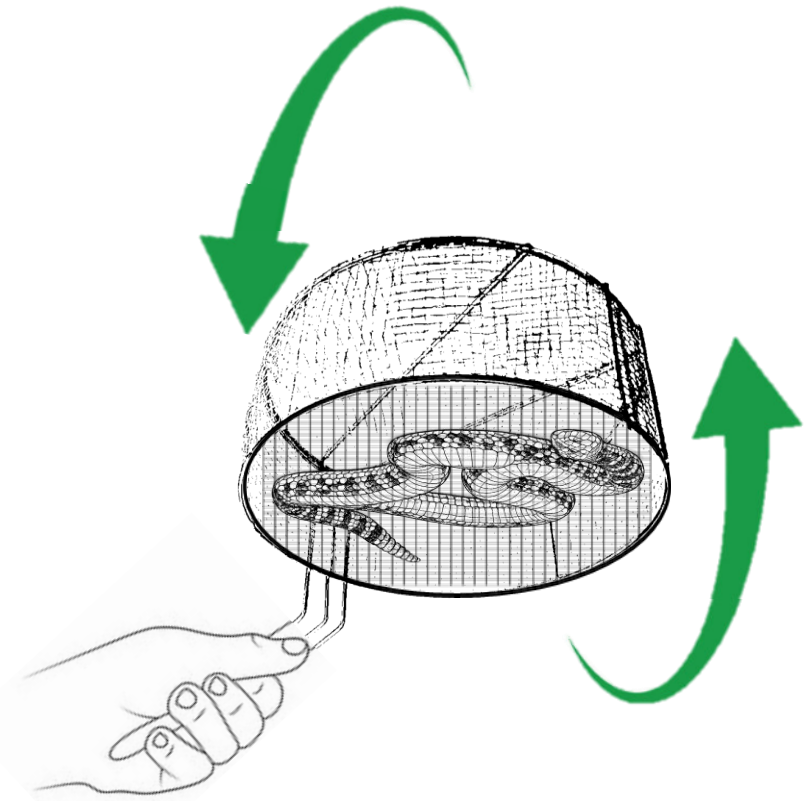




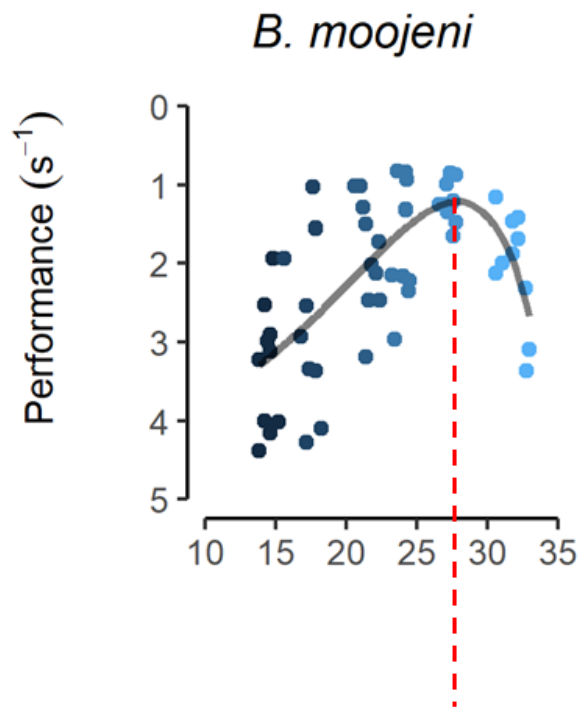


Rãs Manteigas (*Leptodactylus* spp.)

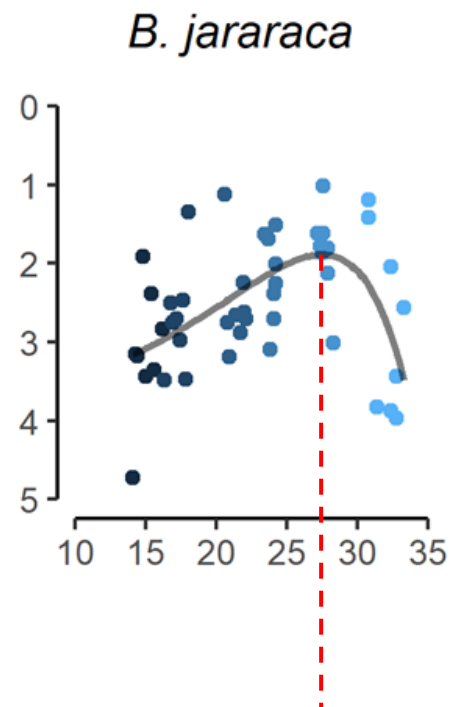




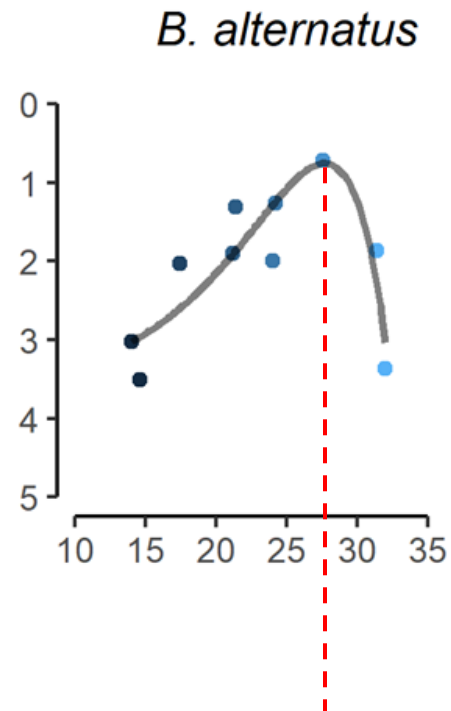




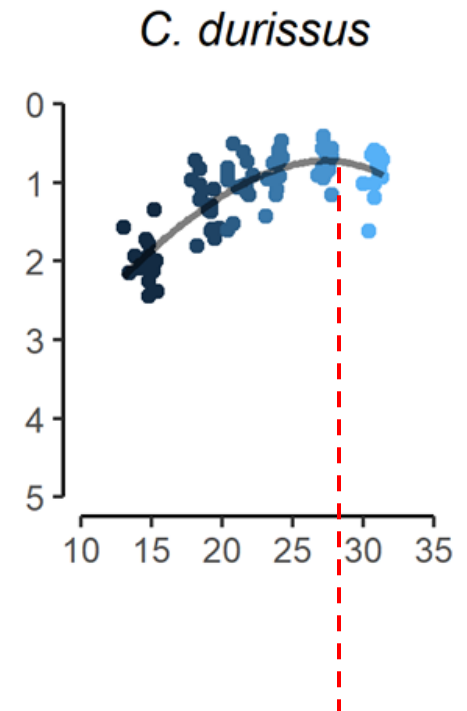
CTmin = 6.8°C
CTmax = 38.9°C



CTmin = 4.7°C
CTmax = 39.7°C



CTmin = 1.9°C
CTmax = 41.3°C

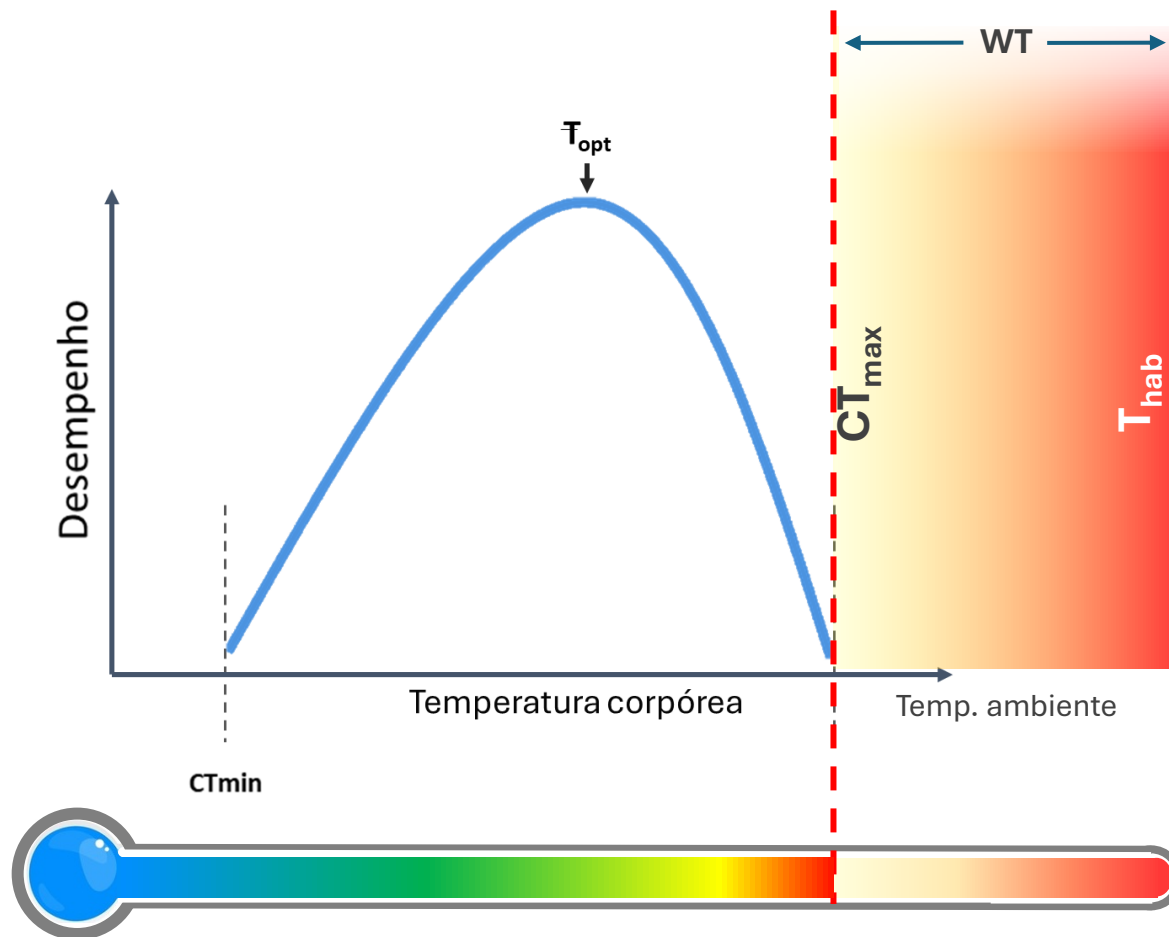


CTmin = 5.5°C
CTmax = 41.8°C

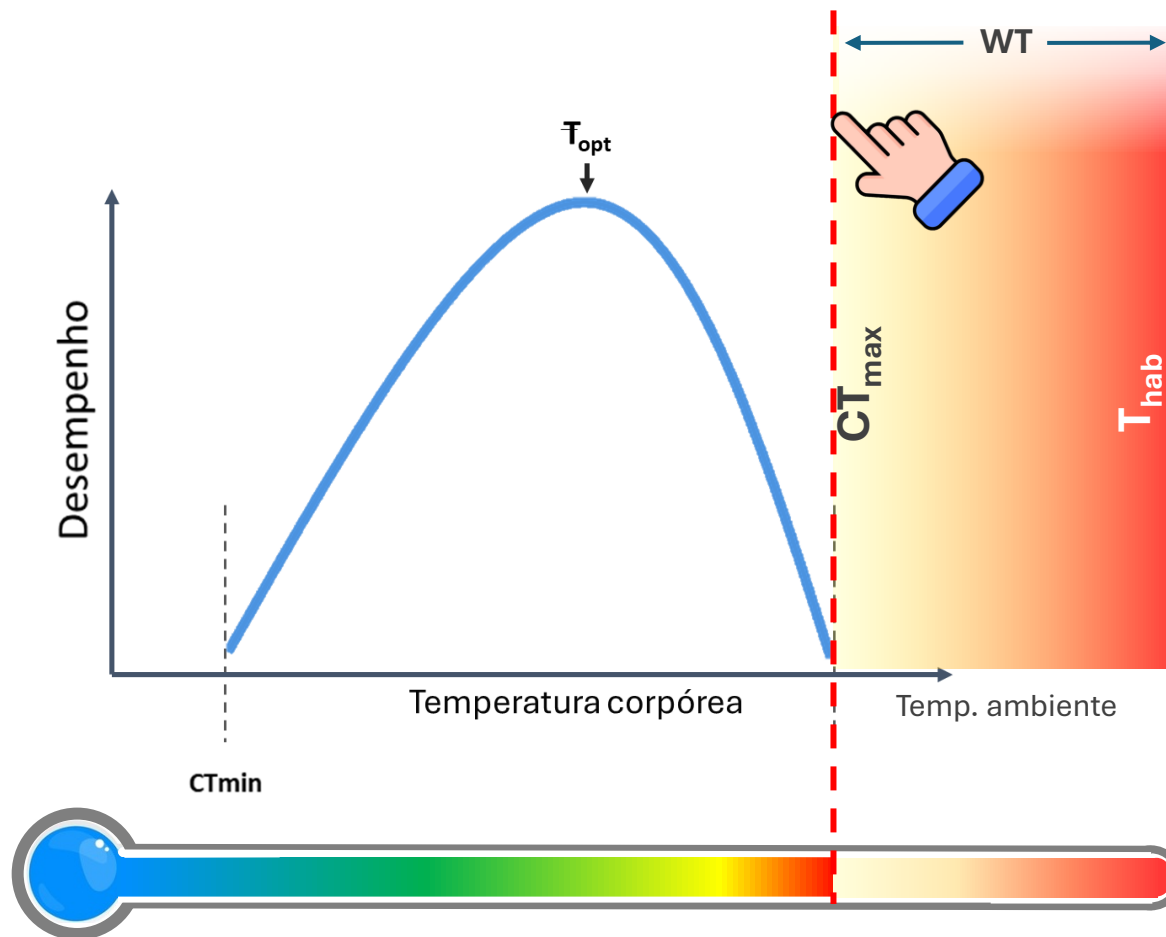
WARMING TOLERANCE (WT)

Deutsch et al. 2008

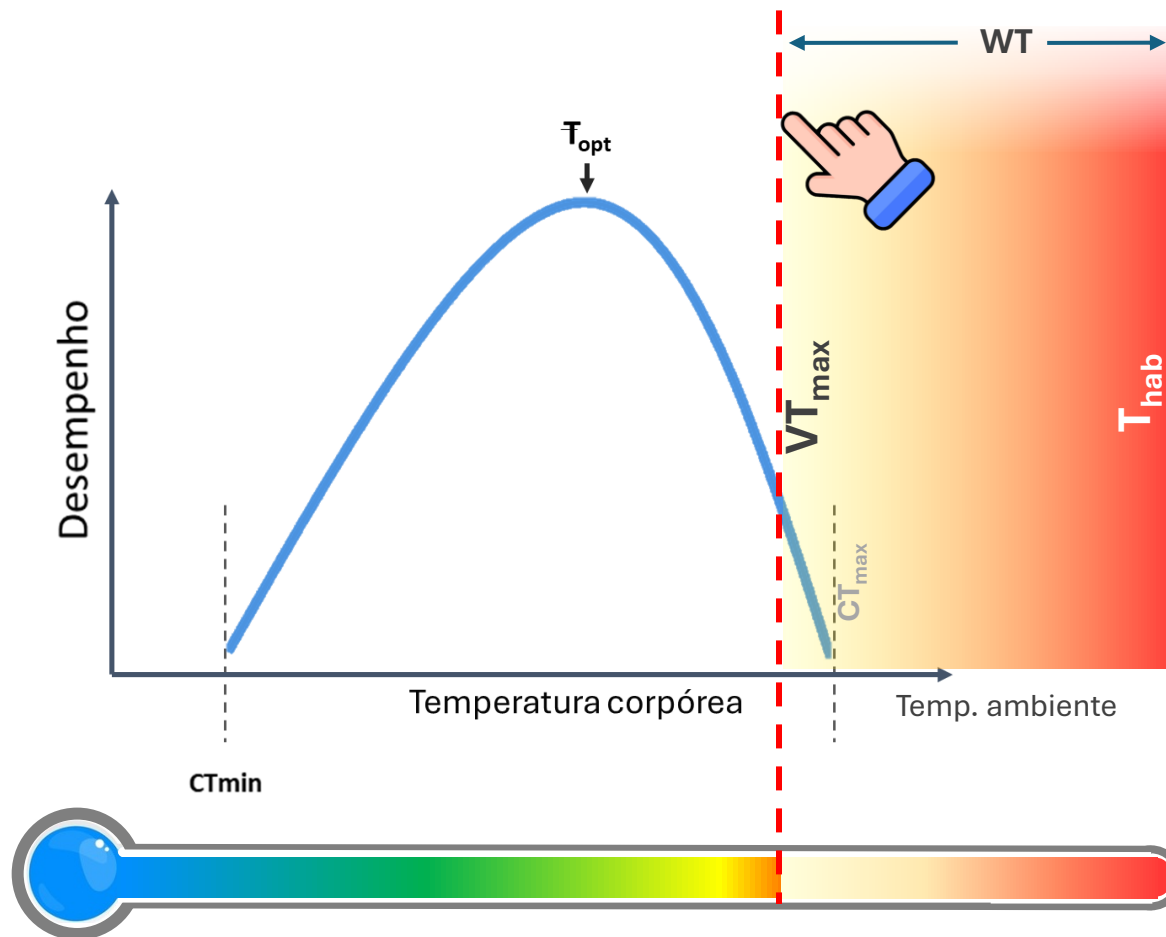
WARMING TOLERANCE (WT)



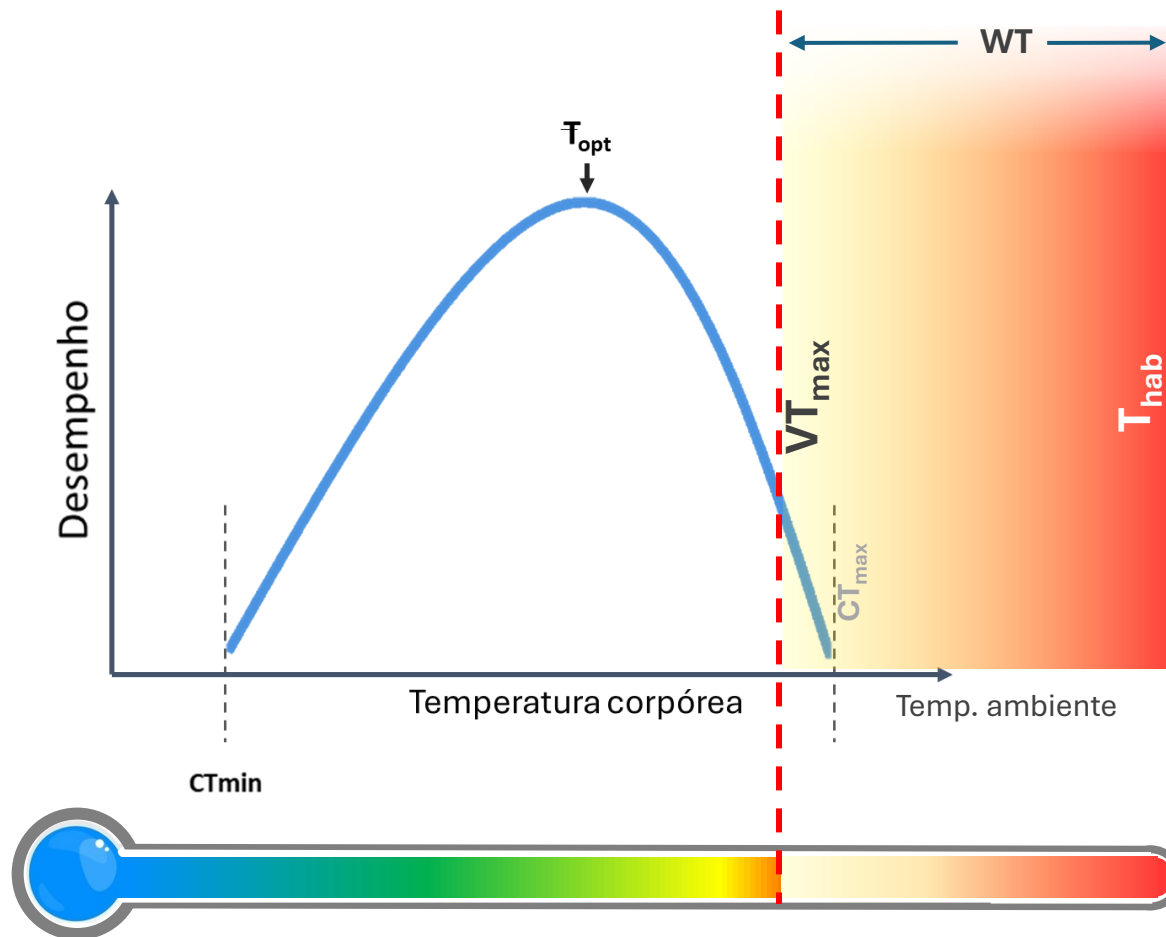
WARMING TOLERANCE (WT)



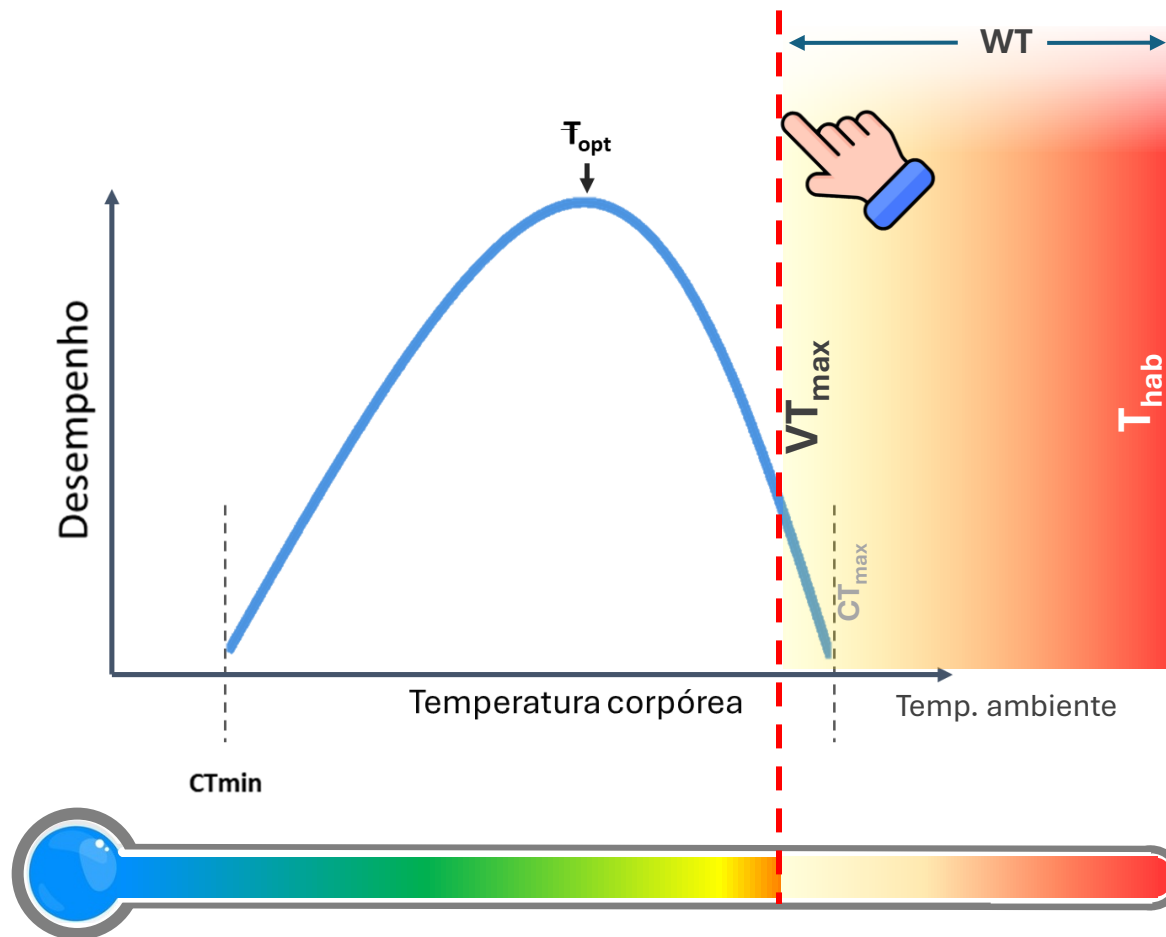
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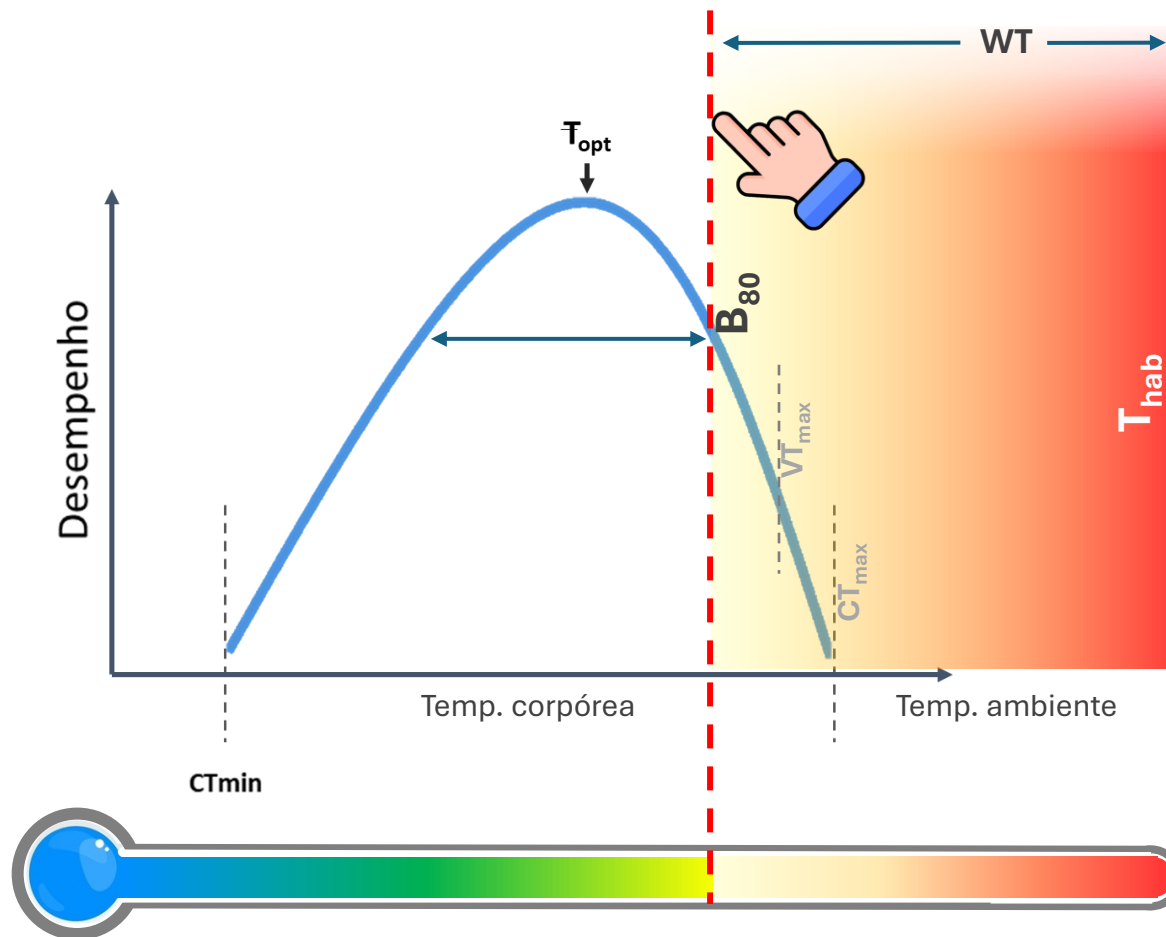
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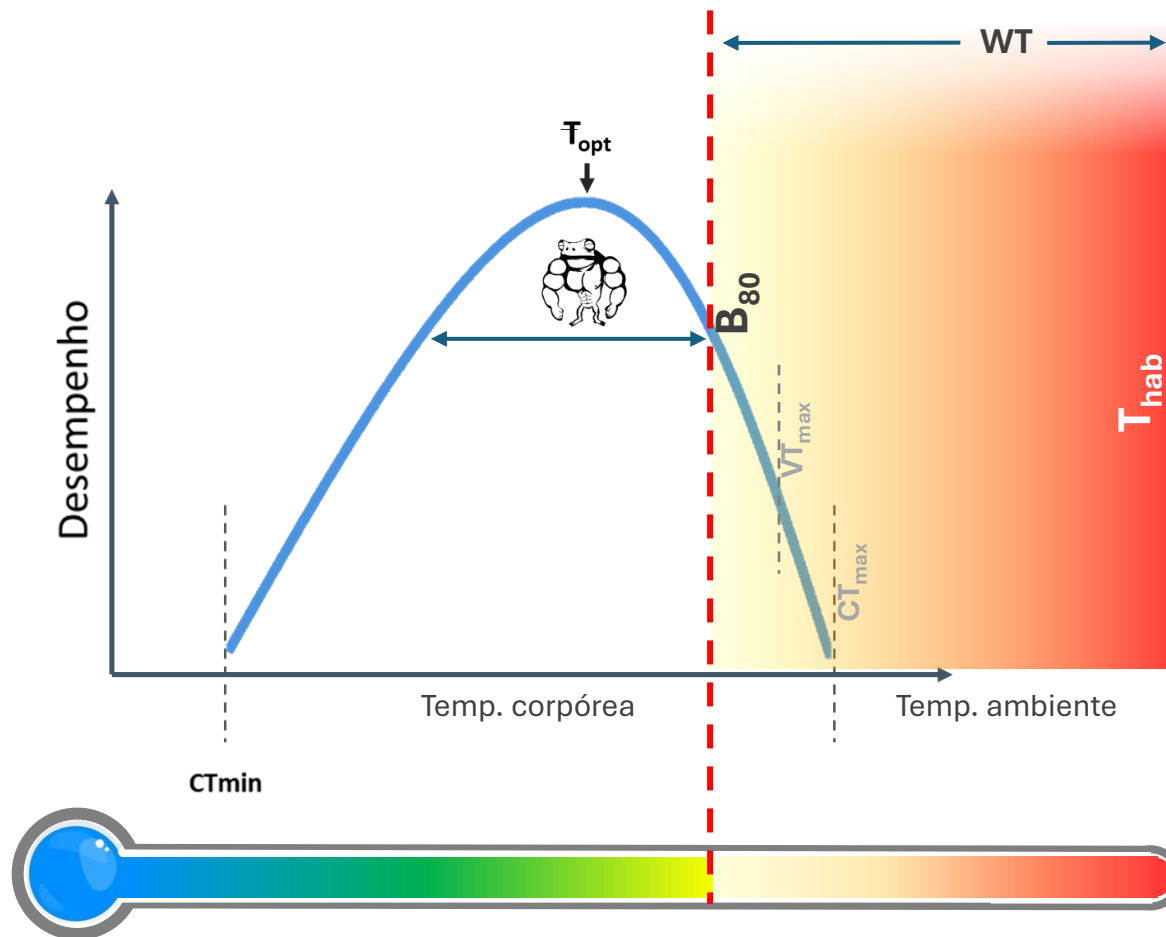
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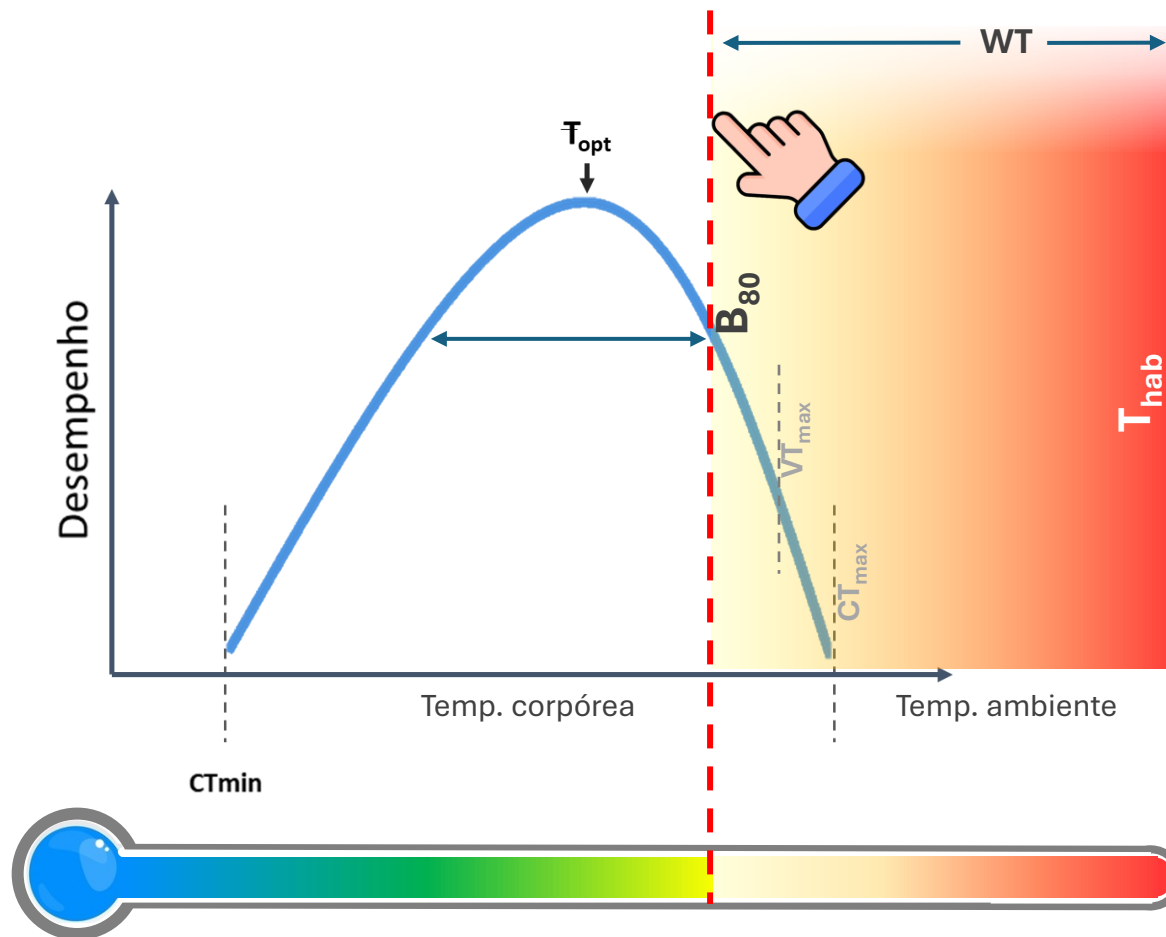
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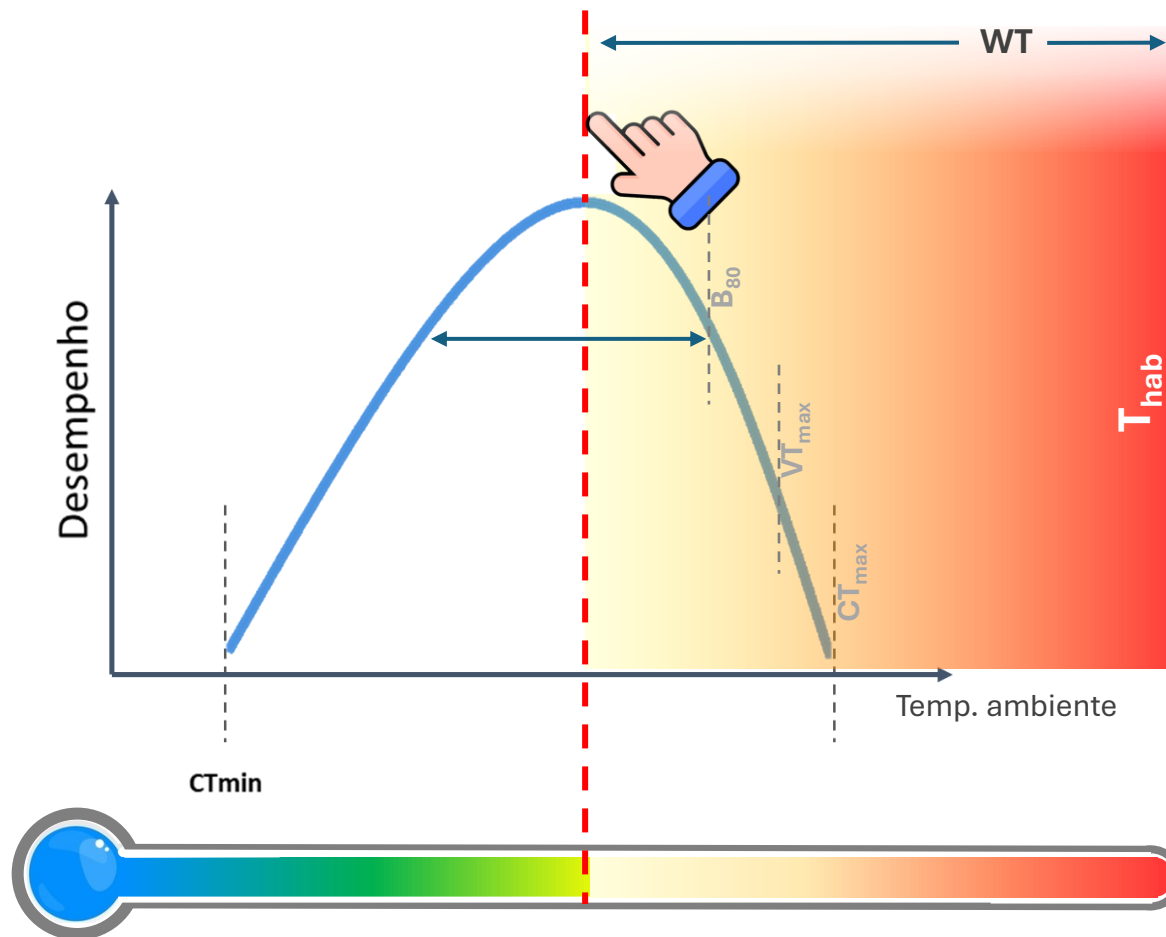
WARMING TOLERANCE (WT)



WARMING TOLERANCE (WT)



WARMING TOLERANCE (WT)



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